

Drift Detection (Data Drift vs Concept Drift)

1 What is Drift?

When an ML model is trained, it *learns from patterns in historical data*.

But over time, **real-world data changes** — new behaviors, new trends, new conditions.

When this happens, your model's predictions slowly become **less accurate**.

That gradual change in input data or target patterns is called **drift**.

In short:

Drift = When the world changes, but your model hasn't.

2 Why Drift Happens

Here are common reasons:

- **Customer behavior evolves:**
e.g., shopping trends change (COVID-19 lockdowns made online purchases spike).
- **Sensor/measurement errors:**
e.g., a camera's brightness changes over time or a sensor starts misreading values.
- **External conditions:**
e.g., new laws, currency value changes, or weather shifts affecting patterns.
- **Data collection updates:**
e.g., a new data source is added, or schema changes (columns added/removed).

When any of these occur, the model trained earlier starts making poorer predictions.

3 Types of Drift

There are **two main types** of drift you must know:

Data Drift and **Concept Drift**.

A. DATA DRIFT (a.k.a. Covariate Shift)

Definition:

When the **distribution of input features (X)** changes over time, but the **relationship between X and Y** remains the same.

→ The inputs look different from what the model saw during training.

Example:

A loan approval model trained on data where “average income = 70k,” but over time, due to inflation, average income becomes “120k.” The model’s decision thresholds are now outdated, even if the rules are the same.

Detection:

Compare feature statistics between training and current data:

- Mean, variance, min, max
- Histogram shifts
- Statistical tests (e.g., Kolmogorov–Smirnov test)

Mitigation:

- Monitor feature distributions regularly.
 - Recalibrate preprocessing or retrain model periodically.
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B. CONCEPT DRIFT

Definition:

When the **relationship between input (X) and target (Y)** changes. In other words, the *concept* your model learned is no longer valid.

→ The same input now produces a *different correct output*.

Example:

An email spam classifier trained in 2022 learned that “free money” = spam.
By 2025, scammers change wording → “urgent opportunity.”
Model fails because **the concept of spam has evolved**.

Detection:

- Sudden drop in model accuracy or F1-score
- Drift metrics: Population Stability Index (PSI), KL Divergence
- Periodic ground truth checks

Mitigation:

- Retrain using new labeled data
- Online learning models (continuous updates)
- Use monitoring pipelines (e.g., SageMaker Model Monitor, Vertex AI Drift Detection)

4 Other Types of Drift (less common)

Type	Description	Example
Label Drift	Distribution of target (Y) changes	Fraud rates increase from 5% → 15%
Feature Drift	Only specific features drift	“Age” column increases, but “income” stays stable
Model Drift	Model degrades due to outdated parameters	Same weights, new world patterns

5 Symptoms of Drift

- Drop in accuracy, precision, recall

- Unusual model outputs
- Unexpected feature importance changes
- Increased prediction latency (if retraining triggers fail)
- Alerts from monitoring tools

6 Tools for Drift Detection in Cloud Platforms

Platform	Tool / Service	Function
AWS	SageMaker Model Monitor	Monitors feature stats & sends drift alerts
GCP	Vertex AI Model Monitoring	Compares live vs training distributions
Azure	Data Drift Monitor	Tracks changes in input data or model outputs

These run automatically and can trigger retraining pipelines.

7 Drift Detection Workflow (Step-by-step)

- 1 **Collect live data** from deployed model.
- 2 **Compare with training data** distributions.
- 3 **Compute drift metrics:** PSI, KL Divergence, KS Test, JS Divergence.
- 4 **Trigger alerts** when thresholds exceeded.
- 5 **Retrain model** or **update feature transformations**.
- 6 **Re-deploy** updated model via CI/CD.

8 Example of Population Stability Index (PSI)

Used to measure how much a feature's distribution changes over time.

PSI Value	Interpretation
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< 0.1	No drift
0.1 – 0.25	Moderate drift
> 0.25	Significant drift, retraining needed

Formula:

$$\text{PSI} = \sum (\text{Actual\%} - \text{Expected\%}) * \ln(\text{Actual\%} / \text{Expected\%})$$

9 Real-world Example

A ride-sharing app's fare prediction model was trained before COVID.

After lockdowns, ride frequency dropped drastically.

Input features like "time of day" and "pickup distance" changed distribution.

➡ The model's fare predictions became inaccurate.

Solution → detected via **data drift**, retrained model with post-COVID data.

10 Why Drift Detection is Critical

Benefit	Explanation
Reliability	Keeps models accurate over time
Compliance	Detects bias changes (important for fairness)
Automation	Enables auto-retraining pipelines
Cost efficiency	Avoids wasted computation on failing models

15 MCQs (with answers + explanations)

1 What does "data drift" mean?

→ Input feature distributions change, but the output relationship stays the same.

2 What does "concept drift" mean?

→ The relationship between inputs and outputs changes over time.

3 Which is more dangerous — data drift or concept drift?

→ Concept drift (it breaks model logic).

4 Which metric measures drift between two distributions?

→ Population Stability Index (PSI).

5 If PSI = 0.3, what does it mean?

→ Significant drift, retraining required.

6 Which AWS service detects drift automatically?

→ SageMaker Model Monitor.

7 Which Azure service detects data drift?

→ Data Drift Monitor.

8 Which GCP tool can detect drift?

→ Vertex AI Model Monitoring.

9 What is the main symptom of concept drift?

→ Drop in model accuracy or F1-score.

10 Data drift affects:

→ Input features (X).

11 Concept drift affects:

→ The relationship between X and Y.

12 Label drift refers to:

→ The target variable's distribution changing.

13 Feature drift means:

→ Only certain features have shifted in distribution.

14 Which test can detect data drift?

→ Kolmogorov–Smirnov test.

15 What's the best way to fix concept drift?

→ Retrain model with fresh labeled data.



10 Short Answer Questions (2–3 sentences each)

1 What is drift in ML?

It's when data or relationships change over time, causing a model's performance to degrade.

2 Difference between data drift and concept drift.

Data drift = change in input distribution; Concept drift = change in input-output relationship.

3 Give one example of concept drift.

Email spam filters stop working because scammers use new keywords.

4 Give one example of data drift.

A housing model fails when average house prices double compared to training data.

5 How can we detect drift automatically?

Use monitoring tools like SageMaker Model Monitor or compute PSI between current and training datasets.

6 What does $PSI < 0.1$ mean?

No significant drift — the model is still stable.

7 Why is drift monitoring essential in production?

Because even a perfect model at training time can fail later as the world changes.

8 What actions should be taken after detecting drift?

Re-examine data preprocessing, retrain with updated data, redeploy.

9 What happens if drift is ignored?

Model accuracy declines, leading to wrong predictions and business losses.

10 How does CI/CD help in drift management?

It allows automatic retraining and redeployment when drift alerts are triggered.